

Tokyo Stock Exchange "arrowhead"

FLEX Market by Order Specifications

(Operational Version 1.1)

Adopted from November 20, 2024

Tokyo Stock Exchange, Inc.

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1 Introduction

These specifications set out details regarding the outline of service, message specifications, operations, requirements, and other aspects of the "Market by Order" type market data dissemination service adopted by the cash equity trading system, arrowhead.

Please use the MBO service and these specifications upon agreement with the following "Terms and Conditions of Use".

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8. User, if applicable, shall enter into the necessary agreements with each exchange for receiving information from the exchange.
9. TSE shall reserve the right to provide the exchange(s) with each FLEX User's join status for multicast groups that contain information from the exchange(s).

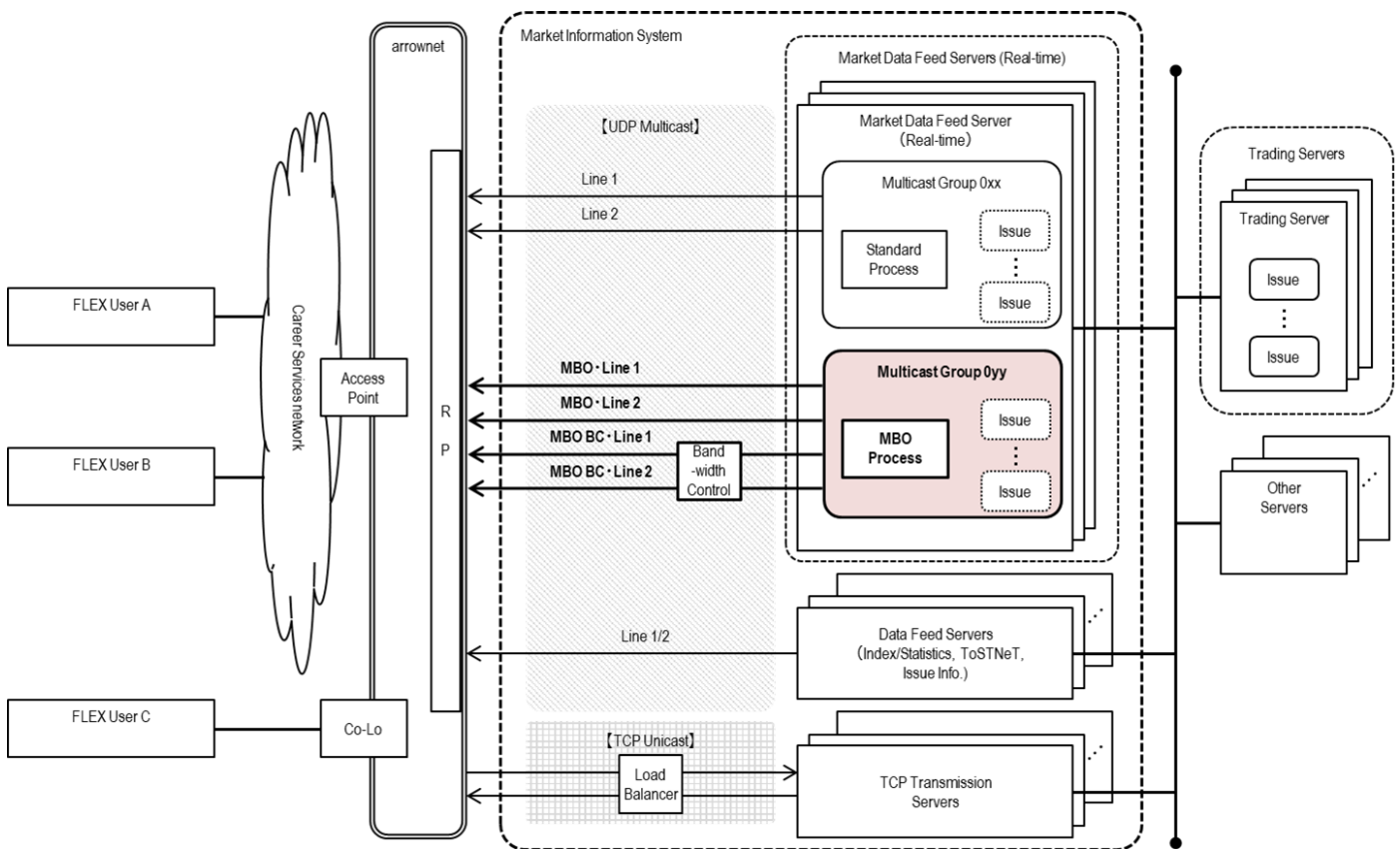
2 TSE Market by Order Service

2.1 Outline

- TSE FLEX Market by Order (MBO) is a service which distributes market data on a by-order basis.
- This enables users to grasp information from the order book in more detail than from a conventional market data service.
- In addition, the MBO service mainly uses binary data to reduce data traffic in order to distribute with high speed and high frequency.
- TSE also provides an "MBO BC*" service, which has a restriction on data traffic, to enable use of other market data services on the same line.
*BC means 'Bandwidth Controlled'.
- The MBO service is not covered by the measures set out for market information in TSE's "Contingency Plan with Regard to Trading in the TSE Market".

2.2 Architecture

- The architecture for the MBO service is as follows.



(1) Trading Server

- A Trading Server matches orders received from Trading Participants.
- Trading Servers match orders for each issue separately.
- Three Trading Servers make up a cluster. Each Trading Server cluster is tasked with matching orders for multiple issues.
- Each business day, different issues are assigned to each Trading Server cluster for order processing. (The assigned issues remain the same for the specific Trading Server cluster throughout the day.)
- The three Trading Servers in each cluster continuously copy data among each other to promptly recover such data in the event one of the Trading Servers in the cluster malfunctions. This mechanism allows for the FLEX service to continue without moving up 'Number of System Reboots'.
- In the event that all three Trading servers in a cluster go out of order, TSE will halt trading of the issues assigned to that specific Trading Server cluster.

(2) A DF (Market Data Feed) Server for MBO

- A DF Server distributes quotes and price information received from Trading Servers for FLEX.
- DF servers distribute information to each multicast group separately.
- Three DF servers make up a cluster, and each DF Server cluster is tasked with distributing information to multiple multicast groups.
- Each business day, different issues are assigned to each multicast group for order processing. (The assigned issues remain the same for the specific multicast group throughout the day.)
- The three DF Servers in each cluster continuously copy data among each other to promptly recover such data in the event one of the DF Servers in the cluster malfunctions. This mechanism allows for the FLEX service to continue, but **the Number of System Reboots is incremented**.
- In the event that all three DF servers in a cluster go out of order, TSE will halt trading of the issues within the multicast group assigned to those DF servers in accordance with its contingency plan.
- In MBO BC, band-width control mechanism is implemented outside of DF servers.

(3) TCP Transmission Server

- A TCP Transmission Server receives requests from FLEX users to retransmit FLEX messages, etc.
- There are several TCP Transmission Servers in operation to allow for continuation of TCP Transmission even if one of the TCP Transmission Servers malfunctions.
- If all TCP Transmission Servers go out of order, then TCP transmission will be unavailable.

* Difference between sets of issues associated with Trading Servers vs. Multicast Groups

- Under the current system architecture, there are more Trading Servers than DF Servers. This means that each DF Server receives information from multiple Trading Servers.
- Therefore, data transmission pertaining to sets of issues in a Multicast Group (assigned to DF Servers) does not occur in the same exact order as data processing in Trading Servers.
- Note, however, that data transmission/processing will not be out of sequence for a single unique issue

2.3 Event Schedule

- The event schedule of the MBO service for each day is as follows.

Time	Event	Notes
07:00	Multicast Routing Maintenance Messages start	Multicast routing maintenance messages are distributed at one-minute intervals.
07:10	Communication Start messages sent	L tags are distributed to signify the start of communication.
	Health Check starts	L tags are distributed as health checks at one-minute intervals.
	Retransmission/Snapshot requests start	
07:38	Issue information (for the trading day)	
07:40	Base Price information (for the trading day)	
	Multicast Group information	
08:00	Morning session order acceptance starts [TSE/NSE]	
08:30	Morning session order acceptance starts [FSE/SSE]	
09:00	Morning session starts [TSE/NSE/FSE/SSE]	
11:30	Morning session ends [TSE/NSE/FSE/SSE]	
12:05	Afternoon session order acceptance starts [TSE/NSE]	
12:15	Afternoon session order acceptance starts [FSE/SSE]	
12:30	Afternoon session starts [TSE/NSE/FSE/SSE]	
15:25	Pre-closing starts [TSE/NSE/FSE/SSE]	
15:30	Afternoon session ends [TSE/NSE/FSE/SSE]	
17:35	Issue information (for the NEXT trading day)	
17:37	Base Price information (for the NEXT trading day)	
18:00	Multicast Routing Maintenance Messages end	
	Health Check ends	
	Communication End messages sent	L tags are distributed to signify the end of communication.
	Retransmission/Snapshot requests end	

3 Message Specifications for UDP Multicast

3.1 Protocol

3.1.1 Message Format

- The protocol for multicast distribution is UDP/IP.
- The packet distributed consists of an Ethernet header, IP header, UDP header and tag data.
- The tag data consists of a packet header from "[3.2.1 Packet Header](#)" and a tag or tags from "[3.2.2 Tag Types](#)".

Name	Offset	Length	Value	Notes	Example
Packet Header	0	26	Char/Bn		
Tag Length (1)	26	1	Bn	Set as length of first tag.	
Tag Data (1)	27	variable	Char/Bn		
...					
Tag Length (n)	variable	1	Bn	Set as length of n-th tag.	
Tag Data (n)	variable	variable	Char/Bn		

The number of tags in a packet



-The number of tags in a packet is determined by the maximum amount of data that can be delivered. In the current tag specifications, we expect the maximum to be 119 tags (T tag and 118 D tags). Information in one tag is never divided into different packets.

- The values used in tag data are as follows.

Value	Description	Main examples
Char	Indicates a character code. ASCII; left-aligned.	Tag Type, Issue Code, Side, etc.
Bn	Indicates a numerical value encoded by big-endian binary. Unsigned integer type.	All numerical items except Price
Bn (Price)	Indicates an unsigned long long type encoded by big-endian binary. The last four digits are decimal fractions.	Price

3.1.2 Tag Blocking

- When multiple tags are generated at the same time, these tags are blocked together into a packet.
- Since there is an upper limit on the number of tags that can be blocked, if the number of tags generated is over this limit, the packet will be split.
- When a packet is split, whether there is another packet or not can be identified by "Utility Flag".
- Since tags are blocked independently for each issue and each process, data for different issues are never mixed and data for different process, even for the same issue, are never mixed in the same packet.

3.2 Tag Data Types

3.2.1 Packet Header

Name	Offset	Length	Value	Notes	Example
Multicast Group Number	0	1	Bn	- Set to the Multicast Group Number.	52
Number of System Reboots	1	1	Bn	- Set to the number of times the system has been rebooted.	0
Sequence Number	2	4	Bn	- Assigned from 1 on a multicast group basis. In principle, the numbers are consecutive, but if a single DF server failure or the system is rebooted, they are assigned from 1 again. - The combination with Number of System Reboots is unique through each business day.	10030501
Issue Code	6	12	Char	Set to the Securities Identification Code as assigned by the Securities Identification Code Committee. For FLEX MBO, Issue Code is set with left-align. (For FLEX Standard, with right-align.) However, in the case of a Reset tag (R tag) and a Communication Control (L tag), this will be set to a space.	8697
Update Number	18	4	Bn	- Assigned from 1 on a per-issue basis for each business day. In principle, the numbers are consecutive, but if the system is rebooted, they are assigned from 1 again. - Even when a packet is split, the same value will be assigned to information which are generated by one transaction of a Trading Server. - The combination with Number of System Reboots is unique through each business day. However, in the case of a Reset tag (R tag) and a Communication Control (L tag), this will be set to '0'.	6000207
Packet Number	22	1	Bn	- Always set to '0'.	0
Total Number of Packets	23	1	Bn	- Always set to '0'.	0
Utility Flag	24	1	Bn	(For UDP multicast, except for Automatic sending of snapshots) When tag data which are generated by one transaction of a Trading Server cannot be distributed in a single packet, whether there is another packet following or not can be identified by this flag. When there is another packet: set to '1'. Other than the above: set to '0'.	0
Message Count	25	1	Bn	Set to the number of tags in the packet.	20

Note: Please refer to '[5.2 Notes for Receiving Snapshot Information](#)' for the values set in packet headers for snapshot information.

3.2.2 Tag Types

- Tags supporting the MBO service are as follows.

Type	Name	Length	Notes
T tag	Seconds Timestamp	5	- Provides information about the time of processing on the Trading Server to the nearest second. - The time to the nearest microsecond is set in the time field of each tag. - Always delivered at the time the information is generated(Except in the case of delivering a Reset tag (R tag) and a Communication Control (L tag)). - Only one tag is delivered per process. If the packet is split, it is stored at the top of the first packet.
O tag	Trading Status	18	- Provides information about the status of trading on a per-issue basis. - Delivered when there is a change in the tag.
K tag	Execution Summary	46	- Provides summary information about execution. - Delivered when an execution occurs.
A tag	Add Order	26	- Provides information about book registration. - Delivered when a new order or modification order is registered.
E tag	Order Executed	20	- Provides information about registered orders in case of Zaraba execution. - Delivered when an execution occurs in Zaraba.
C tag	Order Executed with Price	29	- Provides information about registered orders in case of Itayose execution. - Delivered when an execution occurs by the Itayose method or an execution occurs while an SQ or STQ is displayed.
D tag	Order Delete	11	- Provides information about order deletion from the order book. - Delivered when an order is deleted via a cancel order or a modification order or when a registered order expires.
R tag	Reset	2	- A tag requesting the resetting (i.e., cancellation) of all distributed information. - When the user receives this tag, the user must clear all valid A tags.
L tag	Communication Control	3	- Provides information about communication control. - Delivered when communication starts and ends, and in between at minute intervals.
II tag	Issue Information	125	Please refer to 'Common Items' and 'Issue Basic Information' in the FLEX specifications.
BP tag	Base Price Information	68	Please refer to 'Common Items' and 'Issue Basic Information' in the FLEX specifications.
MG tag	Multicast Group Number Information	16	Please refer to 'Common Items' and 'Issue Basic Information' in the FLEX specifications.

3.2.3 Tag Formats

(1) Seconds Timestamp – T tag

T tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'T'.	T
Time (Seconds)	1	4	Bn	Set to UNIX time (seconds since 1970-01-01 00:00:00UTC).	1704068110

(2) Trading Status – O tag

O tag						
Name	Offset	Length	Value	Notes	Example	
Message Type	0	1	Char	Set to 'O'.	O	
Time (Microseconds)	1	4	Bn	Indicates the time to the nearest microsecond.	323234	
Market Status	5	1	Bn	Indicates market status.	12	
				11: Morning session order acceptance time		21: Afternoon session order acceptance time
				12: Morning session trading hours		22: Afternoon session trading hours
				-		23: Afternoon session pre-closing
				19: End of morning session		29: End of afternoon session
Status Flag	6	2	Char	Indicates trading status.	Space Space	
				A0: Trading halted (accepting orders)		
				A1: Trading halt lifted		
				B0: Itayose		
				B1: Itayose ended		
				C0: Trading suspended		
				C1: Trading suspension lifted		
				D0: Trading halted (not accepting orders)		
				Space: Other than the above		
Short Selling Status	8	1	Bn	Indicates application of the Short Selling Regulation. 0: Short Selling Regulation is NOT applied. 1: Short Selling Regulation is applied.	1	
Pricing Method	9	1	Bn	Indicates method of pricing or status of quote display after the tag is delivered.	4	
				1: Itayose method		
				2: Zaraba method		
				3: Sell Special Quote displayed		
				4: Buy Special Quote displayed		
				5: Sell Sequential Trade Quote displayed		

O tag					
Name	Offset	Length	Value	Notes	Example
				6: Buy Sequential Trade Quote displayed	
				0: Other than the above (e.g., at the timing of session end)	
Book Center Price	10	8	Bn (Price)	Set to the book center price while the Itayose method is in place or while a quote is displayed.	1050000

(3) Execution Summary – K tag

K tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'K'.	K
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Triggered Side	5	1	Char	Indicates the side whose order triggered the execution.	B
				S: Sell order	
				Space: Itayose execution	
				B: Buy order	
Total Volume	6	6	Bn	Set to the total volume of executed orders (number of shares).	10000
Total Invalidation	12	6	Bn	Set to the total volume of invalidated orders (number of shares).	200
Last Price	18	8	Bn (Price)	Set to the last price at the time the execution was processed.	1000000
Match ID	26	4	Bn	An identifier for each issue is assigned on an execution basis. It is unique and sequential through each business day.	601257
Best Offer	30	8	Bn (Price)	Set to the best offer at the time the execution processing finishes. At market close, it is set to '0'.	1020000
Best Bid	38	8	Bn (Price)	Set to the best bid at the time the execution processing finishes. At market close, it is set to '0'.	980000

(4) Add order – A tag

A tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'A'.	A
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Order ID	5	4	Bn	An identifier for each issue is assigned on an order basis. It is unique and sequential through each business day.	1001054
Side	9	1	Char	Indicates buy or sell. S: Sell order B: Buy order	S
Quantity	10	6	Bn	Set to the order quantity registered on the order book (number of shares).	2000
Price	16	8	Bn (Price)	Set to the price registered on the order book. For market orders, this will be set to the maximum value (64bit).	1050000

A tag					
Name	Offset	Length	Value	Notes	Example
Order Condition	24	1	Bn	Indicates the order condition. 0: Non-conditional 2: On-open 4: On-close 6: Funari	0
Modification Flag	25	1	Bn	When the A tag is generated by a modification order, this indicates whether there is a change of time priority or not. 1: Modification order that does NOT change its time priority (reducing the number of ordered shares, etc.) 0: Cancel order or modification order that changes its time priority	1

(5) Order Executed – E tag

E tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'E'.	E
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Order ID	5	4	Bn	Set to the Order ID that was assigned to the executed order. When a partial execution occurs, the same Order ID may be set multiple times.	1001054
Side	9	1	Char	Indicates sell or buy. S: Sell order B: Buy order	B
Volume	10	6	Bn	Set to the executed volume (number of shares).	100
Match ID	16	4	Bn	An identifier for each issue is assigned on an execution basis. It is unique and sequential through each business day.	601257

(6) Order Executed with Price – C tag

C tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'C'.	C
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Order ID	5	4	Bn	Set to the Order ID that was assigned to the executed order. When a partial execution occurs, the same Order ID may be set multiple times.	1001054
Side	9	1	Char	Indicates sell or buy. S: Sell order B: Buy order	B
Volume	10	6	Bn	Set to the executed volume (number of shares).	100
Match ID	16	4	Bn	An identifier for each issue is assigned on an execution basis. It is unique and sequential through each business day.	601257
Execution Price	20	8	Bn (Price)	Set to the execution price.	1080000

C tag					
Name	Offset	Length	Value	Notes	Example
Adopted Pricing Method	28	1	Bn	Indicates the pricing method in place when the execution occurred. 1: Itayose method 2: Quote displayed	1

(7) Order Delete – D tag

D tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'D'.	D
Time (Microseconds)	1	4	Bn	Set to the time to the nearest microsecond.	323234
Order ID	5	4	Bn	Set to the Order ID that was assigned to the cancelled order.	1001054
Side	9	1	Char	Indicates sell or buy. S: Sell order B: Buy order	S
Modification Flag	10	1	Bn	When the D tag was generated by a modification order, this indicates whether there is a change of time priority or not. 1: Modification order that does NOT change its time priority (reducing the number of ordered shares, etc.) 0: Cancel order, modification order that changes its time priority, or expired order	1

(8) **Reset - R tag**

R tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'R'.	R
Start-End Flag	1	1	Bn	When TSE needs to reset all distributed data, this indicates the start or end of the process. 1: Start (set when dissemination of the reset information is started) When the user receives this tag, the user must clear all valid A tags received prior to that point. 2: End (set when dissemination of the reset information is completed)	1

(9) **Control - L tag**

L tag					
Name	Offset	Length	Value	Notes	Example
Message Type	0	1	Char	Set to 'L'.	L
Test Mode Flag	1	1	Bn	Indicates production or test. 1: Production 2: Test	1
Start-End Flag	2	1	Bn	Indicates the start or end of communication. 1: Communication Start 2: Communication End 0: Health Check control	1

3.2.4 **Tag Priorities**

- When multiple tags are generated by the same process, tag priorities are, in principle, as follows.

Priority	Tag Types	Notes
1	T tag	- Always delivered at the time information is generated.
2	O tag	- Delivered when there is a change in the tag. If there is no change, it is omitted.
3	K tag	- Delivered when an execution occurs. If there is no execution, it is omitted.
4	D tag	- Provides information about order deletion through modification orders or cancel orders, etc.
5	E tag/ C tag, D tag	- Provides information about execution or invalidation. On the same price level, execution (E tag or C tag) comes before invalidation (D tag). When information occurs on multiple price levels, for example when a buying-up occurs, the order of distribution is E tag, D tag, E tag, and so on.
6	A tag	- Provides information about registered order volumes stemming from new orders and modification orders.

Priorities among C tags

-In case of delivering C tags due to normal Itayose execution occurrence, the priorities among C tags are as follows.

(1st) Orders executed at the price at which there are remaining orders(either buy or sell),

(2nd) Buy orders at the price from lower to market buy order, (3rd) Sell orders at the price from higher to market sell order.

3.3 Appendix

3.3.1 Multicast Routing Maintenance Message

- A Multicast Routing Maintenance message is delivered for each multicast group.
- The message doesn't have a packet header, and 1-byte spaces '0x20' are set in the data area.
- The purpose of the message is to maintain and establish a multicast distribution tree, and to prevent message duplication before delivering the control messages (Communication Start and Health Check control), by sending it to the same multicast address as other messages.
- Although the message is distributed from the exchange-side system to users, it is not necessary for the user-side application to capture the message, as long as it reaches the user-side system.

3.3.2 Special Examples to Consider for User-side Applications

(1) On-close orders/Funari orders at and after the start of Pre-closing

- In a session with a closing auction, On-close orders are brought onto the order book as valid orders after Pre-closing starts.
- In a session with a closing auction, Funari orders are brought onto the order book, not as limit orders but as market orders, after Pre-closing starts.
- At the time Pre-closing starts, information about On-close and Funari orders being brought onto the order book as above is not distributed.
- Also, On-close orders and Funari orders are sometimes sent during Pre-closing.
- If a user constructs an order book or calculates indicative closing prices based on information from the MBO service, it is necessary to consider the above examples on the user-side system.

(2) A value set in 'Utility Flag' tag in packet header in a UDP multicast

- In opening auctions or other Itayose auctions, a large amount of execution information is generated.
- When tag data which are generated by one matching process cannot be distributed in a single packet, the packet is split.
- Also, in order to deliver an Execution Summary message without delay, a packet that contains a K tag is sometimes distributed prior to other packets.
- In the above situations, by setting the [Utility Flag](#) value to '1: When there is another packet', it is possible to identify whether the information was generated by the same matching process or not.
- The same method is used to deliver messages in the case of Zaraba execution where there is a large number of orders executed.
- 'Packet Number' and 'Total Number of Packets' are set to '0' at the time of arrowhead4.0 operation.

(Distribution example) Itayose execution in an opening auction

Packet No.	Total No. of Packets	Utility Flag	Example of Tags	Notes
0	0	1	T O K C C ... C	In order to deliver the K tag without delay, the first packet is delivered before the other packets. Since the Utility Flag is set to '1', there will be more messages that were generated by the same matching process.
0	0	1	C C C ... C C	Since the Utility Flag is set to '1', there will be more messages that were generated by the same matching process.
:	:	:		
0	0	1	C C C ... C C	Since the Utility Flag is set to '1', there will be more messages that were generated by the same matching process.
0	0	1	C C C ... C D	Since the Utility Flag is set to '1', there will be more messages that were

Packet No.	Total No. of Packets	Utility Flag	Example of Tags	Notes
				generated by the same matching process.
0	0	0	D D D . . . D D	The last packet

(3) **Example of where 'Price' in an A tag is replaced, e.g., when the initial price of a newly listed issue is determined**

- Before the initial price of a newly listed issue is determined, the price limit range is not prescribed since there is no base price. When an initial price is determined during a session, the upper/lower limit prices will then be determined based on that initial price. Given this, On-close buy orders that had been placed at a price higher than the new upper limit price and On-close sell orders that had been placed at a price lower than the new lower limit price will be treated as orders placed at the upper/lower limit price.
- When a base price is changed after order acceptance starts, buy orders that had been placed at a price higher than the new upper limit price and sell orders that had been placed at a price lower than the new lower limit price will be treated as orders placed at the upper/lower limit price.
- Since [A tags](#) will not be redistributed in the above situations, it is necessary to consider this on the user-side system.

4 Abnormal Operations

4.1 Retransmission and Snapshot via TCP

4.1.1 Outline

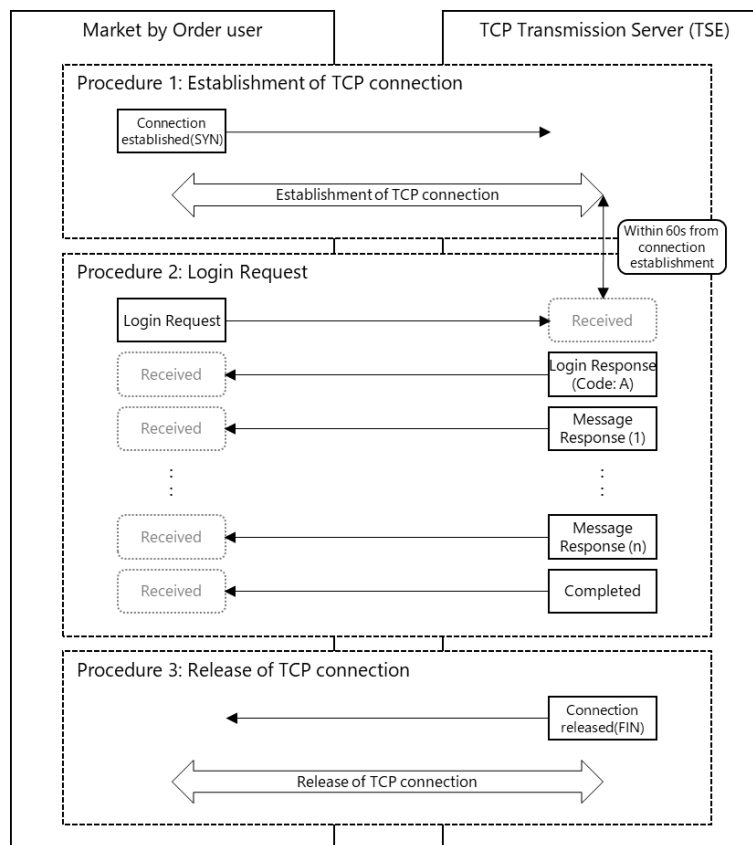
- A retransmission function and snapshot request function are provided via TCP in order to recover messages after a message drop is detected via UDP multicast on the user-side system or after recovery from a failure on the user-side system.
- The retransmission function is a function that requests messages to have been distributed via UDP multicast.
- The snapshot function is a function that delivers valid orders that were on the order book at the time of the request as [A tags](#).
- When using these TCP message request functions, please understand the contents of this chapter and strictly comply with '[7 Compliance](#)'.

4.1.2 Message Format

- The protocol of the retransmission and snapshot functions is TCP/IP.
- The packet consists of an Ethernet header, IP header, TCP header and tag data.
- The tag data consists of a packet header from '[3.2.1 Packet Header](#)' and a tag or tags from '[3.2.2 Tag Types](#)'. The structure of the packet is [the same as for UDP multicast](#).

4.1.3 TCP Request Flow

■ Normal transaction



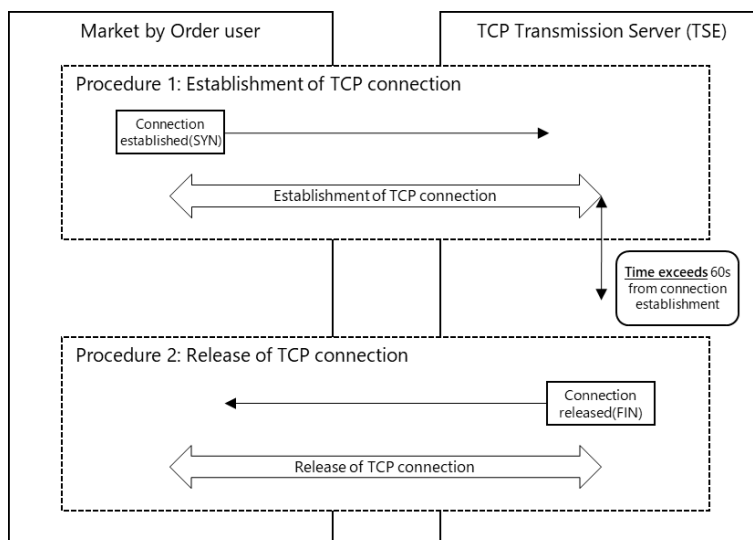
- Users should establish a TCP connection from a fixed IP address.
- After establishing a TCP connection, please send a request message. The User ID prescribed in the specifications must be indicated at this time.
- The exchange-side system will release the TCP connection after sending an 'End of Message' message.



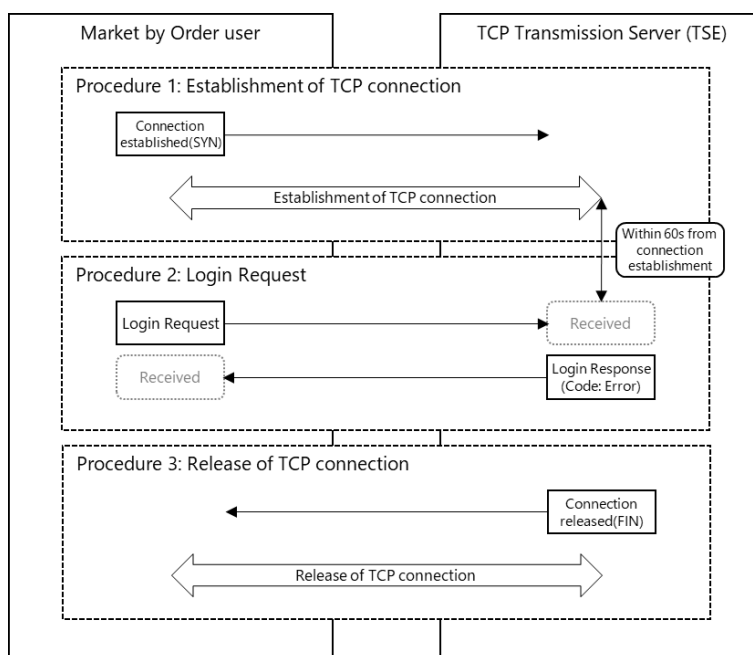
In case where there are no available messages for response

If the retransmission data is not ready when the request message is received for the information delivered by UDP multicast, the exchange-side system sends a login response message (Code: A), but then releases the TCP connection.

■ Error transaction



- After establishing a TCP connection, if the exchange-side system doesn't receive a request message within a fixed time, the TCP connection will be released from the exchange-side system.



- If the request is normal, messages will be sent in response. If the request contains incorrect information, the exchange-side system will send an error response and release the TCP connection.



Notes on disconnecting the TCP connection

For both normal and error processes, the TCP connection should be released by sending a FIN from the exchange-side system. Users should refrain from disconnecting the TCP connection with a FIN or RST.

4.1.4 Message Type

Type	Source	Length	Note
Login Request	User	18	This is the same for retransmission requests and snapshot requests.
Login Result	arrowhead	5	
Message Response	arrowhead	Variable	
End of Message	arrowhead	9	

(1) Login Request

Login Request					
Name	Offset	Length	Value	Notes	Example
Packet Length	0	2	Bn	Set to the message length excluding the digits included in 'Packet Length' (this item).	16
Packet Type	2	1	Char	Indicates the packet type. R: Retransmission request L: Snapshot request	R
User ID	3	6	Char	Set to the User ID for login authentication.	TSE101
Multicast Group Number	9	1	Bn	Set to the Multicast Group Number for the request target.	51
Number of System Reboots	10	1	Bn	(Retransmission) Set to the number of times the system has been rebooted. (Snapshot) Set to '0'.	0
Sequence Number	11	4	Bn	(Retransmission) Set to the first sequence number for the request. (Snapshot) Set to '0'.	1
Requested Message Count	15	3	Bn	(Retransmission) Set to the number of messages from the first sequence number for the request. (Snapshot) Set to '0'.	20

User IDs

- For the time being, User IDs will be shared with all users. It is possible for more than one user to connect simultaneously with the same User ID.

User ID: FLEX01 *F/L/E/X/Zero/One

Dummy Symbols

- For Dummy Symbols, only TCP retransmission is available, TCP snapshot is not.

**Maximum number of 'Requested Message Count'**

- Maximum number of 'Requested Message Count' is 50,000.

(2) Login Result

Login Result					
Name	Offset	Length	Value	Notes	Example
Packet Length	0	2	Bn	Set to the message length excluding the digits included in 'Packet Length' (this item).	3
Packet Type	2	1	Char	Indicates the packet type. A: Accepted J: Rejected	J
Multicast Group Number	3	1	Bn	Set to the Multicast Group Number for the request target.	51
Result Code	4	1	Char	Set to the result code of the login request.	A
				A: Accepted	

Login Result					
Name	Offset	Length	Value	Notes	Example
				O: Out of service	T: Incorrect packet type
				U: Incorrect User ID	M: Incorrect MCG number
				S: Incorrect sequence number	Z: Other error

(3) Message Response

Message Response					
Name	Offset	Length	Value	Notes	Example
Packet Length	0	2	Bn	Set to the message length excluding the digits included in 'Packet Length' (this item).	150
Packet Type	2	1	Char	Set to 'S'.	S
Data	3	Variable	-	Tags targeted by the request are sent in the response including their packet headers.	-

(4) End of Message

End of Message					
Name	Offset	Length	Value	Notes	Example
Packet Length	0	2	Bn	Set to the message length excluding the digits included in 'Packet Length' (this item).	7
Packet Type	2	1	Char	Set to 'G'.	G
Multicast Group Number	3	1	Bn	Set to the Multicast Group Number for the request target.	51
Number of System Reboots	4	1	Bn	(Retransmission) Set to '0' (Snapshot) Set to the number of times the system has been rebooted.	0
Next Sequence Number	5	4	Bn	(Retransmission) Set to '0'. (Snapshot) Indicates the next multicast transmission sequence number that should be received after capturing the snapshot information.	1

4.1.5 Differences from UDP Multicast

(1) Retransmission

- There are no differences between retransmission messages and messages distributed through UDP multicast.

(2) Snapshot

- Information is sometimes sent in snapshot messages that is not distributed via UDP multicast since it is the latest information at the time of the request. (For example, if an order is executed partially, the quantity of [A tags](#) delivered as part of the snapshot is the quantity of the new order minus the executed quantity.)

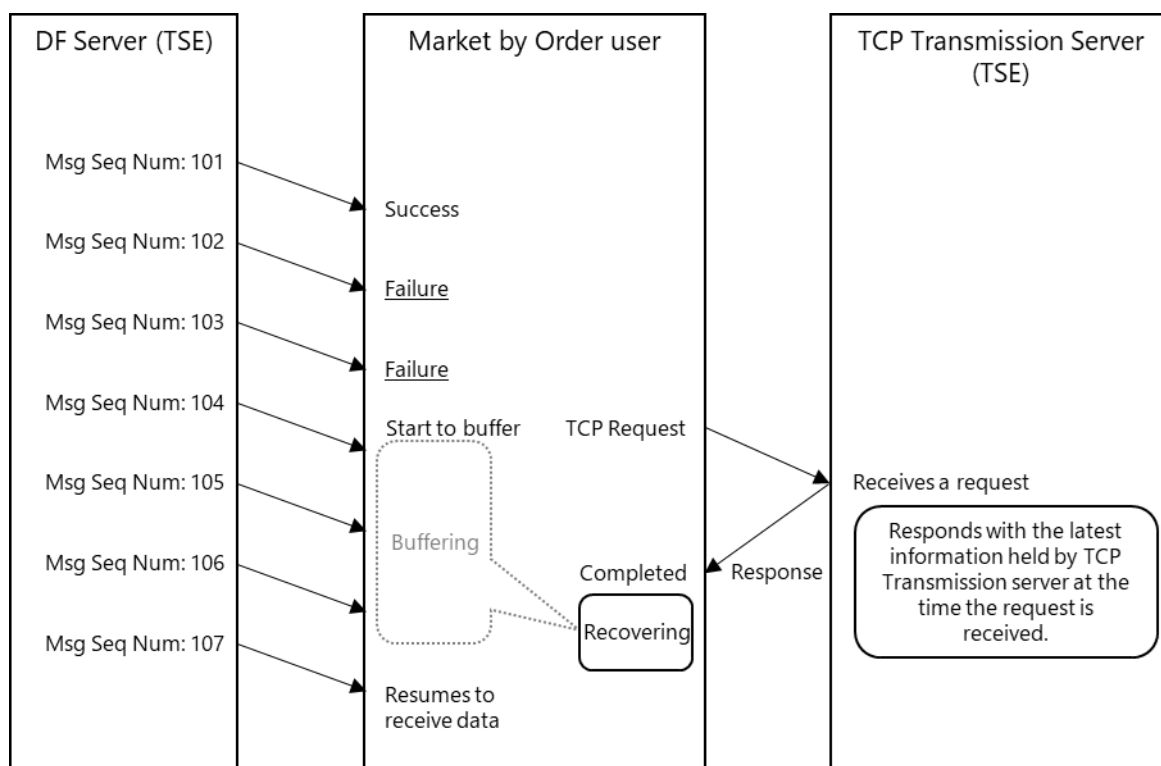
- '[Number of System Reboots](#)' and '[Sequence Number](#)' in [the packet header](#) are set to '0'. After capturing the snapshot information, the next sequence number that should be received via UDP multicast is identified by 'Number of System Reboots' and 'Next Sequence Number' in the End of Message message.

- Snapshot messages consist of a T tag, an O tag, and A tags (the number of valid orders).

- 'Time (Seconds)' in the [T tag](#) is set to the last time that the O tag was updated. 'Time (Microseconds)' in the [O tag](#) is set to the last time the O tag was updated, and 'Time (Microseconds)' in the [A tag](#) is set to '0'.
- For other points to note regarding the snapshot function, please refer to '[5.2 Notes for Receiving Snapshot Information](#)'.

4.1.6 Example of how to recover the latest order books by TCP message request functions

- For recovery using retransmission function and snapshot request function by TCP, if it is used during trading hour after order acceptance start, it is necessary to receive and buffer (hold) the information via UDP multicast since order books (quotes) will change even before the TCP request is completed.
- After the TCP request is completed, users are expected to complement the data with the buffered information and resume to receive data via UDP multicast. An example is as below.



[Commentary]

- Message sequence number (Msg Seq Num) 101 is normally received. Msg Seq Num 102 and 103 are failed due to user-side incident.
- After the error occurred, user send a TCP request. (In case of retransmission, user sets Msg Seq Num 102 to 'Sequence Number' and set any number of messages to 'Requested Message Count'.
- At the same timing of the TCP request, user starts to buffer data via UDP multicast.
- After the TCP request is completed, users recover data with TCP request messages.
(In this case, if user receives information up to Msg Seq Num 103, user complements Msg Seq Num 102 and 103.)
- After recovery by TCP, users recover data with the buffered messages.
(In this case, user complements from Msg Seq Num 104 to 106.)
- User will resume to receive and operate from Msg Seq Num 107.

4.2 Failure in Exchange-side Systems

4.2.1 A single DF server failure

(1) 'Number of System Reboots' increment

- If a failure occurs in one of [the DF servers](#), 'Number of System Reboots' is incremented and distribution continues. In this case, 'Sequence Number' restarts from 1, but 'Update Number' continues consecutively from those before the failure.

(2) Automatic sending of snapshots via UDP multicast

- If a failure occurs in one of [the DF servers](#), as there may have been a message drop, the TSE system will send [R tags](#) (reset tags) before resending all valid order information as A tags.

- The snapshot message consists of a T tag, an O tag, and A tags (the number of valid orders).

- 'Time (Seconds)' in the [T tag](#) is set to the last time the O tag was updated. 'Time (Microseconds)' in the [O tag](#) is set to the last time the O tag was updated, and 'Time (Microseconds)' in the [A tag](#) is set to '0'.

4.2.2 Recovery operations when FLEX messages are not transmitted due to a failure

- This section describes TSE's recovery operations when FLEX MBO messages are not transmitted due to a system failure, etc. When a system failure occurs, users have to prepare to restart on the premise of operations described in this section.

(1) Premise

Users have to consider the following points regardless of TSE's recovery operations.

- Users are required to join both lines of UDP multicast. If users detect a failure, etc. in one line, users have to continue operation by receiving messages from the other line without TSE's announcement.

- If Users detect a temporary message lost, etc. in both lines of UDP multicast, users have to continue operations by retrieving the lost messages using TCP request function.

(2) Policies of recovery operation

- When a system failure, etc. recovers and UDP multicast becomes available, TSE resumes UDP multicast. When TSE resumes UDP multicast, please refer to '[4.2.1 \(2\) Automatic sending of snapshots via UDP multicast](#)'.

- The recovery policies and the expected behaviors for each information are shown in the table below.

Contents	Recovery Policies*		Expected behaviors of recovery in UDP multicast
		Notes	
Multicast Routing Maintenance Message	C	If it is recognized as a failure where only Multicast Routing Maintenance messages are not transmitted and TSE announces this situation to users, TSE may continue to provide this service.	After recovery, Multicast Routing Maintenance messages will start to be transmitted.
Communication Start (L tag)	B	If Health Check messages are normally transmitted, since this situation can be seen that the communication is normal, TSE may continue to provide this service. (For a process of user-side system that is driven by the Communication Start as a trigger, it is preferable to be able to be driven by manual operation.)	As to the message sequence number, while the Communication Start is transmitted with the message sequence number of 1 in each multicast group in a normal operation, the message sequence number at the timing of the recovery will be set in the recovery operation.

Contents	Recovery Policies*		Expected behaviors of recovery in UDP multicast
		Notes	
Health Check Control (L tag)	B	If it is recognized as a failure where only Health Check message is not transmitted and TSE announces this situation to users, TSE may continue to provide this service.	The message sequence number at the timing of the recovery is set and messages will start to be transmitted at regular intervals (every 1 minute).
Communication End (L tag)	C	This message may not be recovered on the day if there is no prospect of early recovery. (For a process of user-side system that is driven by the Communication End as a trigger, it is preferable to be able to be driven by manual operation.)	The message will be transmitted with message sequence number at the timing of the recovery.
Other tags (T/O/K/A/E/C/D)	A	TSE will continue operation without trading halt in case of a single-line failure or a failure of this service.	The latest order book will be recovered by transmitting snapshots over UDP multicast upon recovery. If TSE cannot provide snapshots over UDP multicast, users need to continue operations by requesting snapshots via TCP and recovering the latest order book

*Legend for "Recovery Policies" column

A: Information necessary to continue operation.

B: Information whose recovery is not required to continue operation depending on conditions.

C: Information that may not be recovered because it has no direct impact on continuing operation.

4.2.3 Notes on recovery through a reboot of TSE's system (trading restart)

- This section describes the notes on recovery by rebooting the TSE's system on the day due to unexpected failure of the TSE's system.

(1) Operational Schedule

- When the TSE's system is rebooted, all messages that are described in events of '2.3 Event schedule', that is all information after Multicast Routing Maintenance message, will be distributed. TSE will announce the actual time schedule for the day after the system reboot.

(2) Operation of rolling forward 'Order ID' and 'Match ID'

- For 'Order ID' and 'Match ID', TSE will roll forward the values in order to avoid overlaps of the values between the messages that will be transmitted after rebooting and the messages that had been transmitted before the system failure. The values that will be applied after rebooting will be announced on the day.

(3) **'Number of System Reboots' increment**

- If a failure occurs whose impact on operations is unforeseen, TSE's system may be rebooted to restart trading. In this case also, '[Number of System Reboots](#)' will be incremented, however, both '[Sequence Number](#)' and '[Update Number](#)' restart from 1.

(4) **Clearing the quote information**

- When the TSE's system is rebooted, the quote information that had been transmitted before the system failure is required to be cleared but TSE may not be able to transmit [the Reset information \(R tag\)](#) depending on a failure situation. In this case, in order to restart by rebooting, TSE will ask users to clear the quote information possessing on the user-side system on the notifications.

4.3 Failure in MBO User-side Systems

4.3.1 Provision of Message Data File

(1) Outline

- When MBO user had not received FLEX messages due to the user-side system failure, TSE provides FLEX data in another way according to users' request.
- TSE stores at least last two weeks' FLEX data, but the data may not be available due to system constraints.
- The output messages of Test Environment are not available.

(2) Definition of File

- The file will conform to the FLEX Historical service connection specifications and will be provided as a PCAP file.

(3) Provision of Each Service Data

- The method of data transfer will be determined with MBO user on case-by-case basis.

(4) Data Format

- The data format will conform to the FLEX Historical service connection specifications. Please refer to "Specifications for Connecting to FLEX Historical Service" published by JPX Market Innovation & Research, Inc.

(5) Procedure for Provision of Message Data File

- If MBO user needs data, please contact JPX Service Desk.

5 Examples of Distributed Messages

5.1 Basic Examples

- Please refer to 'Annex 01_Examples of Distributed Messages'.

5.2 Notes for Receiving Snapshot Information

- Packet headers for distributing snapshot information are as below, for both UDP and TCP.

Name	UDP	TCP	Diff.
Multicast Group Number	- Set to the Multicast Group Number.	- Set to the Multicast Group Number.	None
Number of System Reboots	- Set to the number of times the system has been rebooted.	- Set to '0'.	<u>Yes</u>
Sequence Number	- Assigned from 1 on a multicast group basis.	- Set to '0'.	<u>Yes</u>
Issue Code	- Set to the Securities Identification Code prescribed by the Securities Identification Code Committee. - In case of an R tag, this is set to a space.	- Set to the Securities Identification Code prescribed by the Securities Identification Code Committee.	None
Update Number	- Set to the latest Update Number.	- Set to the latest Update Number.	None
Packet Number	- Set to '0'.	- Set to '0'.	None
Total Number of Packets	- Set to '0'.	- Set to '0'.	None
Utility Flag	A snapshot response, depending on the timing, may be a snapshot of the order book in the middle of the process, not reflecting all the information created by one transaction of a matching process (a state in which 'Utility Flag' is set to '1'). In this case, whether there is more information to come or not is identified by this flag. When there is more information: set to '1'. Other than the above: set to '0'.	A snapshot response, depending on the timing, may be a snapshot of the order book in the middle of the process, not reflecting all the information created by one transaction of a matching process (a state in which 'Utility Flag' is set to '1'). In this case, whether there is more information to come or not is identified by this flag. When there is more information: set to '1'. Other than the above: set to '0'.	None
Message Count	Set to the number of tags in the packet.	Set to the number of tags in the packet.	None

6 List of IP Addresses and Port Numbers

6.1 Production Environment

(1) UDP multicast

- Please refer to "Annex02_List of IP addresses and Port Numbers".

(2) TCP retransmission and snapshot

Service	MCG number	Exchange-side		User-side	
		IP address	Port number	IP address	Port number
All	All	10.17.6.200	57500	Any	Any



Issue basic information (II tag, BP tag, MG tag)

- Please refer to the FLEX specifications concerning Multicast Group Numbers, IP addresses and Port Numbers from issue basic information.



For use of base price information (BP tag)

- Base price information is provided as a paid service. For more details, please refer to 'Policies Regarding Usage of Market Information'.

6.2 Test Environment

(1) UDP multicast UDP

- Please refer to "Annex02_List of IP addresses and Port Numbers".

Note: Neither test environment has a distinction between "Unlimited" and "Limited".

(2) TCP retransmission and snapshot

Service	MCG number	Exchange-side		User-side	
		IP address	Port number	IP address	Port number
All of Test Env. A	All	10.17.6.200	57500	Any	Any
All of Test Env. B	All	10.87.133.200	57500	Any	Any

7 Compliance

7.1 Requirements

(1) General items

- This service assumes that the user receives messages via 2 lines (2 circuits).
- If the user detects an abnormal incident on the user-side (a message drop, etc.), it is assumed that the user's system is set up to continue operations by receiving messages through the other line.
- TSE assumes that if user detects an abnormal incident in both lines, for example a temporary message drop, they will use the TCP retransmission function, and if a user recovers from a long-term failure to receive messages and wants to capture the latest information, they will use the TCP snapshot function.
- The service's data circuit terminating equipment depends on each carrier's specifications.
- The service's receiving-side router supports IP multicast. Its IGMP version is version 2, and its protocol is PIM-SM.

(2) Usage of retransmission and snapshot functions

- Since the user is required to connect via 2 lines (2 circuits) to use the MBO service, the retransmission function and snapshot function are assumed to be for use in the case that a message drop is detected in both lines. Even if a user doesn't receive messages on two lines and uses a single line for retransmission and snapshot functions, TCP requests may be restricted if there are problems such as frequent repetition.
- For TCP requests, users are required to comply with the specifications and procedures in [4.1 Retransmission and Snapshot via TCP](#) and to adequately test the user-side system on holidays in the Production Environment and on weekdays in the Test Environment before starting use. For testing, the person in charge of testing on the user side should be clearly identified and the user must ensure that said person is set up to respond promptly to contact from the exchange side on the test day.
- Establishing a TCP connection without actually transmitting the required request message and sending repeated request messages containing errors are strictly prohibited since these actions can overload the TSE servers, thereby affecting other users.
- Only one request is allowed to establish a single connection. If the user sends another request message while the exchange-side system is responding to the first request, the exchange-side system will not respond to the latter request.
- TCP retransmission and snapshot functions are for recovering specific messages that were not received. Retrieving a large number of messages through repeated retransmissions or snapshots is strictly prohibited since it may cause the TSE servers to overload, thereby affecting other users.
- When a TCP request fails, a mechanism is required to be implemented to control requests so that requests with errors are not sent repeatedly.

Revision History

November 5, 2024. New release (Operational Version 1.0)

November 20, 2024. Update (Operational Version 1.1)

- Newly added to information column about TCP Request Flow. [page 15]
- Described precautions when disconnecting TCP connection. [page 16]
- Added the distribution examples of messages when a trade is executed while displaying a Special Quote, etc. [Annex01]